What is OOP?

INTRODUCTION TO OBJECT-ORIENTED PROGRAMMING IN JAVA

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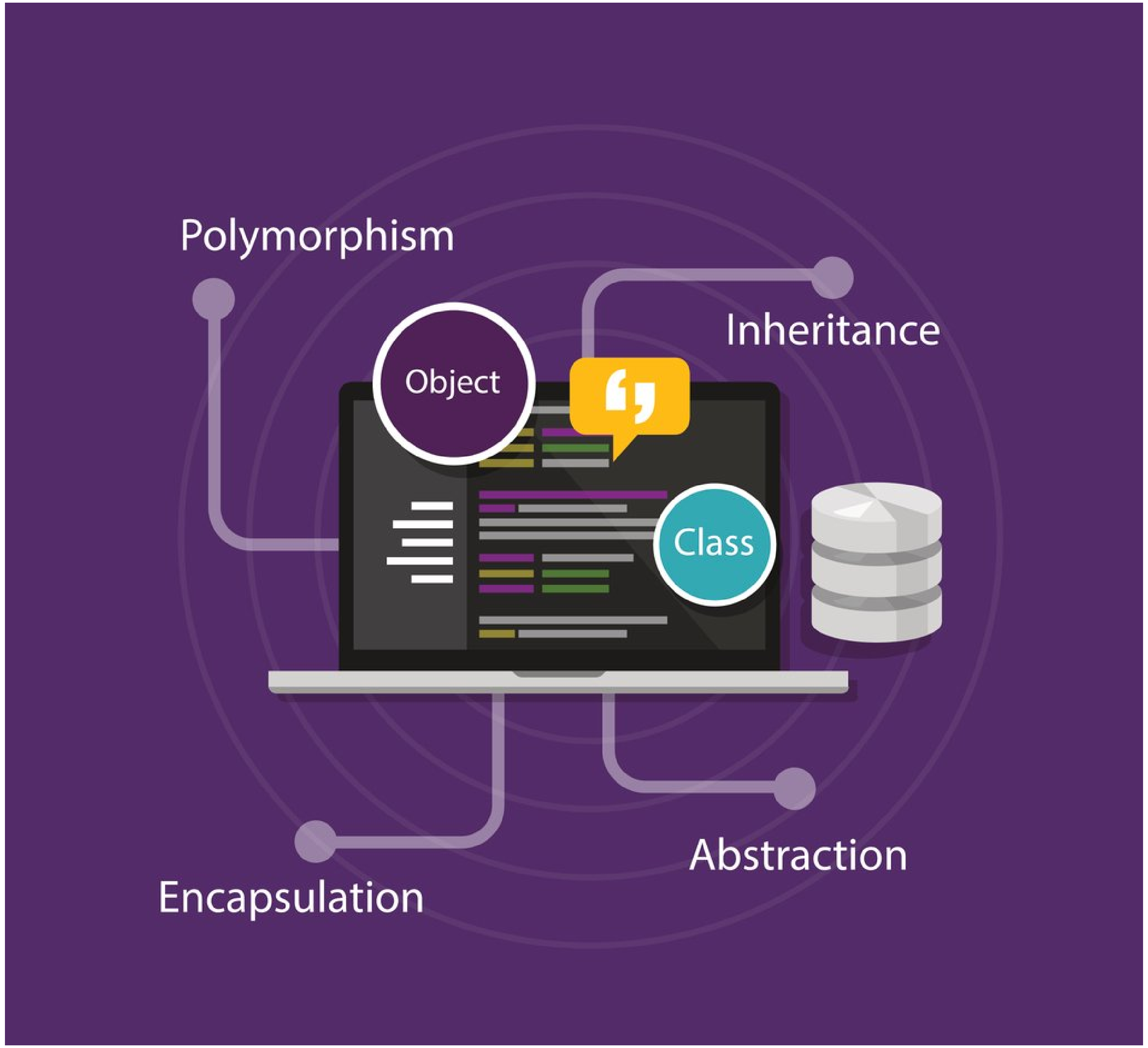
About me

What will we learn?



- Classes versus Objects
- Constructors, fields, and methods
- `new` keyword
- Access modifiers & encapsulation
- Inheritance & interfaces
- Polymorphism

What is Object-Oriented Programming (OOP)?

- Programming paradigm
- Model real-world entities
- Data = Fields
- Code = Methods

```java
public class User {
    String username; // Field
    String password; // Field
    boolean validateCredentials(String username, String password) { // Method
        return this.username.equals(username) && this.password.equals(password);
    }
}
```

Everything is an Object in Java

```java
public class Main {  // A class is required even for the simplest Java program!
    public static void main(String[] args) {
      System.out.println("Hello World!");
   }
}
```

We create objects to fit our needs

```java
public class User {
    String username; // Field
    String password; // Field
    boolean validateCredentials(String username, String password) { // Method
        return ...;
```

Primitive data types

Data Type	Size	Limits
byte	1 byte	Numbers from -128 to 127
short	2 bytes	Numbers from -32,768 to 32,767
int	4 bytes	Numbers from -2,147,483,648 to 2,147,483,647
long	8 bytes	Numbers from -9,223,372,036,854,775,808 to 9,223,372,036,854,775,807
float	4 bytes	Fractional numbers. Sufficient for storing 6 to 7 decimal digits
double	8 bytes	Fractional numbers. Sufficient for storing 15 decimal digits
boolean	1 bit	`true` or `false` values
char	2 bytes	A single character/letter or ASCII values

[1] https://www.w3schools.com/java/java_data_types.asp

Let's practice!

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Classes vs Objects

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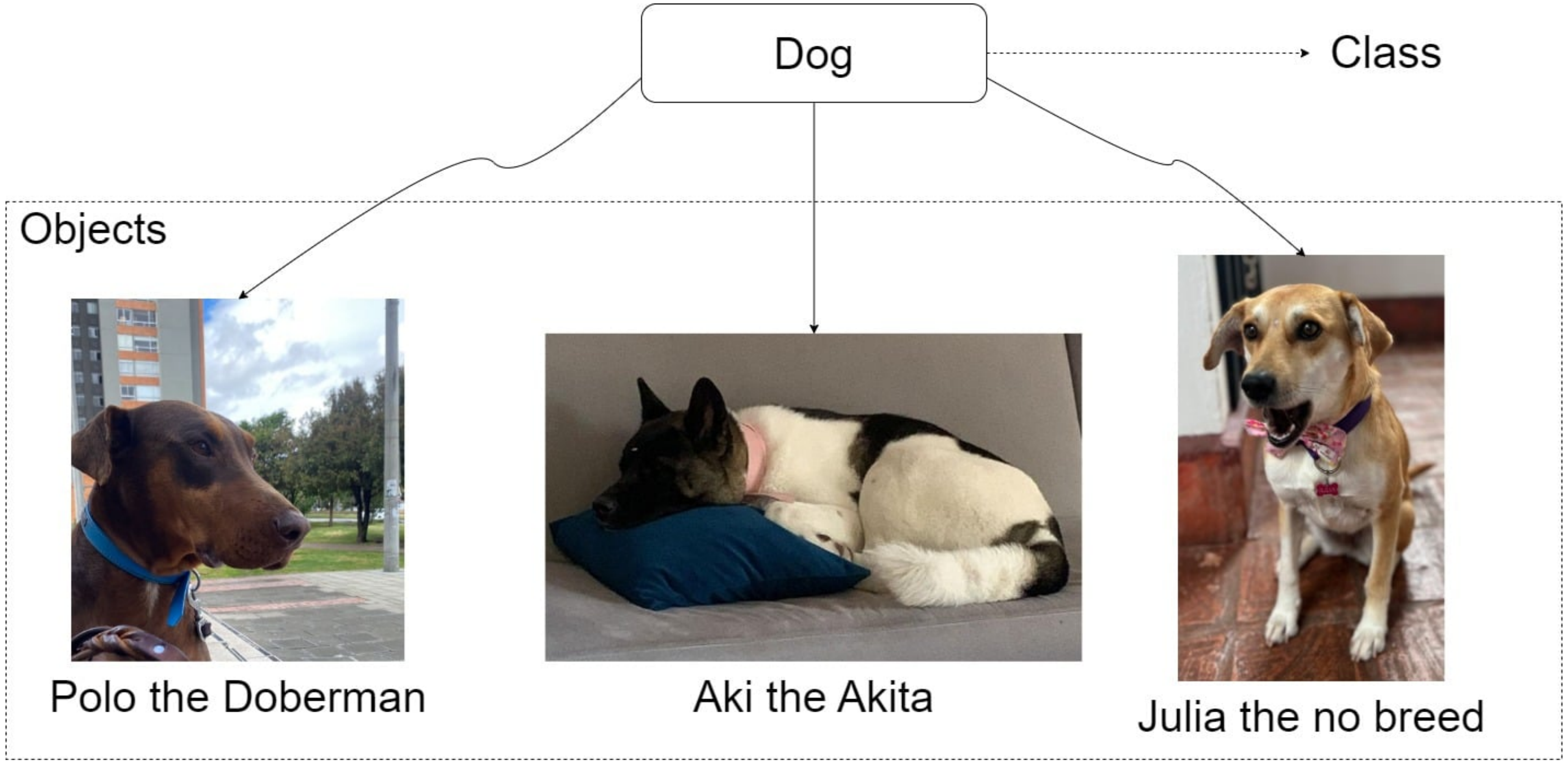
What is an object?

- Real-world entities
- Basic unit in OOP
- Fields to store data, and methods to provide behavior
- A Doberman named "Polo"

What is a class?

- How do we create objects?
- Classes describe how objects will look like
- Classes are blueprints for creating objects



Creating objects from classes

```java
public class Dog {
  // Fields
  String name;
  String breed;

  // Constructor
  Dog(String name, String breed) {
    this.name = name;
    this.breed = breed;
  }
}
```

```java
public class DogDaycare {
  public DogDaycare() {
    // `new` keyword to create objects
    Dog polo = new Dog("Polo",
                       "Doberman");
    Dog aki = new Dog("Aki", "Akita");
  }
}
```

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Components of an object/class: Fields, Constructor, and Methods

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The class body

```java
// Class header
class Dog {
  // Fields
  String name;
  int age;
  String bark;
  // Constructor
  Dog(String name, int age) {
      this.name = name;
      this.age = age;
      this.bark = "Woof!";
  }
}
```

The class body: Methods

```java
class Dog {
  String name;
  int age;
  String bark;
  Dog(String name, int age) {
    this.name = name;
    this.age = age;
    this.bark = "Woof!";
  }
  void bark() { // method
    System.out.println(bark + " name = " + name + ", age = " + age);
  }
}
```

The object

```java
public class DogDaycare {
  Dog polo;
  Dog aki;
  public DogDaycare() {
    // `new` keyword to create objects
    this.polo = new Dog("Polo", 4);
    this.aki = new Dog("Aki", 3);
  }
  void makeDogsBark() {
    polo.bark(); // Output: Woof! name = Polo, age = 4
    aki.bark(); // Output: Woof! name = Aki, age = 3
  }
}
```

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